Web Mining
Assignment 3 – LLM

Laura Raileanu
Cédric Campos Carvalho, Elena Najdenovska

Institut des Technologies de l'Information et de la Communication (IICT)

Introduction

- **“Meet Ollama ! ”**, an introduction to the library *Ollama* and how embeddings work.
- **“Retrieval Augmented Generation (RAG)”**, where you explore how to perform RAG on a model with a new source of data.
- **“Create a chat ! ”**, where you use the library `chainlit` to create a chatbot with RAG for an uploaded file.
- **“Scraping recent news information”**, explore how to perform RAG with data scraped about the recent news.

No report to give, only the files completed.

Deadline : 23th April 2025 at 23:59:59

General Tips

- Read the documentation of the libraries used.
  - Often examples are given
  - `ollama-python` doesn't have a very detailed documentation, but it is a python "interface" for the Ollama API.
- Steps with RAG often look about the same.
- **Docker Desktop** is a great tool if you're not familiar with docker.

- For the notebook, try to clear all cells outputs once and run everything. **Keep the outputs for the upload on Moodle.**

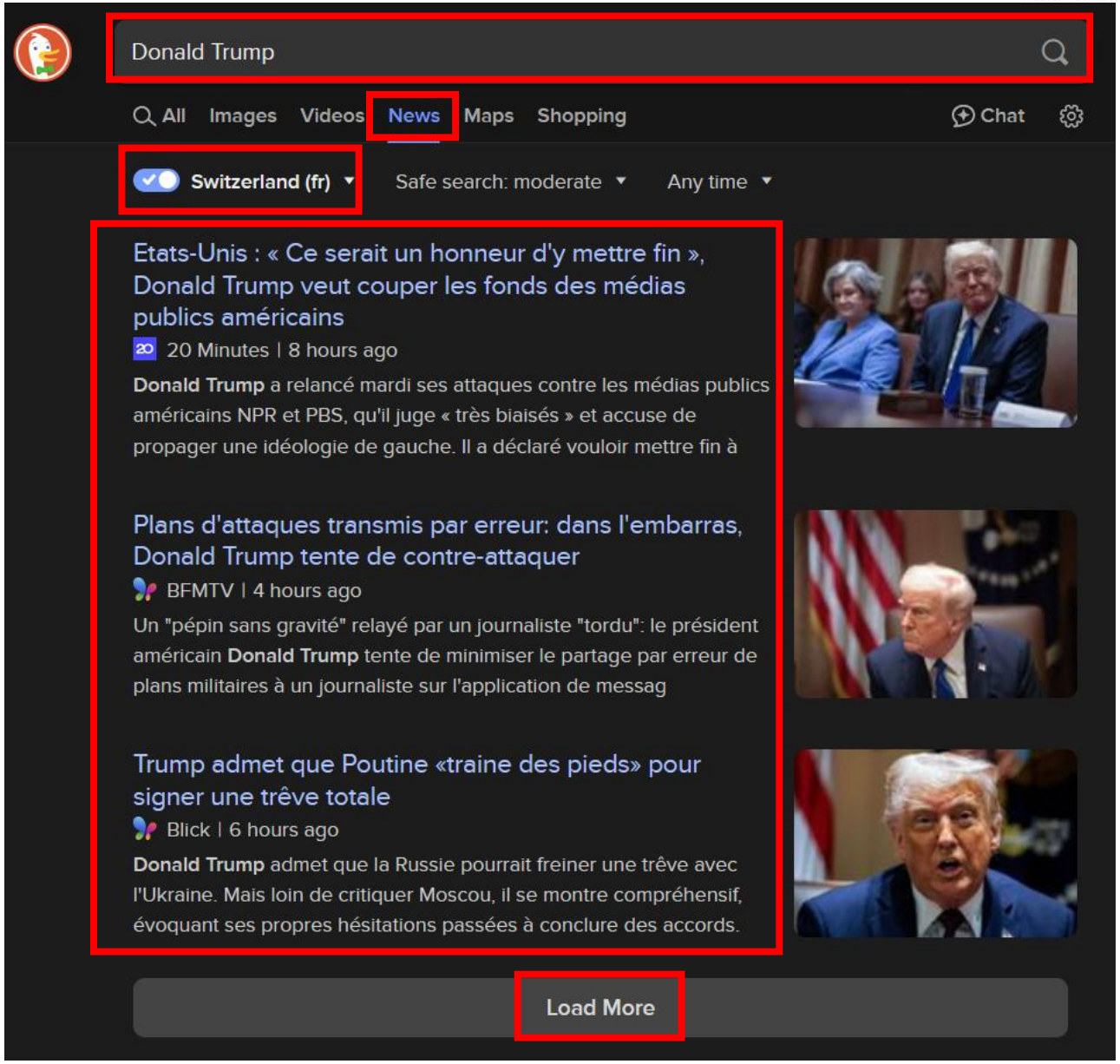
*Exercise 3 - **Create a chat !***

Demonstration

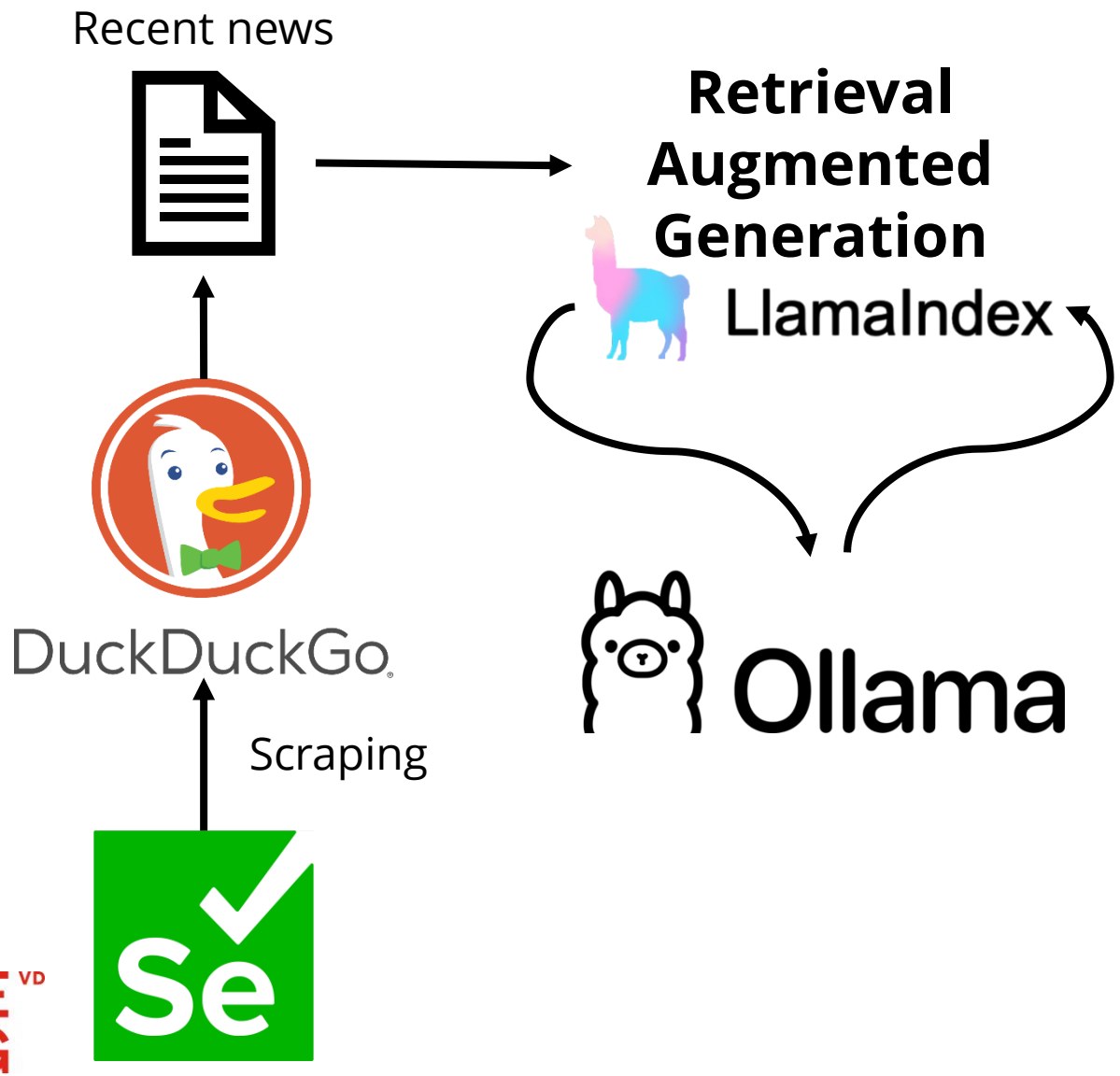
Exercise 4 - Scraping recent news information

- Scrap news information from DuckDuckGo
- RAG with the previous information

Key points for the scraping in **red**.



Exercise 4 - Scraping recent news information




```python
search_term = "JO 2026"
region = "Switzerland (fr)"
context = create_context(search_term, region) # goes scraping
                    ...
print_answer(llm, context, question)
```

Dernières nouvelles sur la thématique JO 2026:

- La Suisse a remporté sa 23e et dernière médaille aux Jeux olympiques de Milan-Cortina, lors d'une cérémonie de clôture qui se déroulera vendredi soir.

- Franjo von Allmen est devenu champion olympique en super-G, tandis que Marco Odermatt a obtenu la médaille de bronze dans cette épreuve.

- Nadja Kälin a remporté une sensationnelle médaille de bronze sur le 50 km en ski de fond.

- Les USA sont champions olympiques et ont battu le Canada 2-1 après prolongation dans une finale pourtant dominée par les joueurs à la feuille d'érable.